

Original Article

Artificial intelligence as a discursive practice: the case of embodied software agent systems

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Abstract In this paper, I explore some of the ways in which Artificial Intelligence (AI) is mediated discursively. I assume that AI is informed by an “ancestral dream” to reproduce nature by artificial means. This dream drives the production of “cyborg discourse”, which hinges on the belief that human nature (especially intelligence) can be reduced to symbol manipulation and hence replicated in a machine. Cyborg discourse, I suggest, *produces* AI systems by rhetorical means; it does not merely describe AI systems or reflect a set of prevailing attitudes about technology. To support this argument, I analyse a set of research articles about an “embodied conversational agent” called the Real Estate Agent (REA). The articles about REA mobilise a set of rhetorical strategies that systematically downplay the system’s artificiality and bolster its humanlike qualities. Within the context of the dream of AI to produce humanlike machines, and given our strong bias for human-human interaction, the designers’ claim to REA’s humanness in their research articles, as I argue in the final section of this paper, needs little justification.

Keywords Discourse · Embodiment · Metaphor · Rhetoric · Software agents

Introduction

Artificial Intelligence (AI) has been defined as the “art of creating machines that perform functions that require intelligence when performed by people” (Kurzweil 1990). To the extent that this art depends, like other kinds of scholarly inquiry (see Latour and Woolgar 1986), upon the production of physical as well as discursive artefacts, AI is mediated by rhetoric. In this paper, I explore some of the ways in which the “ancestral dream” of AI to build a “*homunculus* or thinking machine” (Baggi 2000) is negotiated in discourse. I treat AI as a discursive practice and assume that intelligent machines are scaffolded (Gee 1999) by a constellation of discourses. One such constellation is “cyborg discourse” (see Vaux 2001), which hinges on the belief that human nature (especially intelligence) can be reduced to symbol manipulation and hence replicated in a machine. The driving metaphor of cyborg discourse is: A Computer Is A Human.¹ From a discourse analytic perspective, cyborg discourse neither merely reflects a prevailing or pre-existing attitude about the possibility of thinking machines, nor merely describes objects and ideas that presumably exist beyond language and discourse. Rather, the view espoused in this paper is that cyborg discourse *produces* machines as human rather than simply describes them as such. Cyborg discourse is performative, machines are inscribed in texts (see Latour 1988), and the “presence, meaning, and value” of technologies are always mediated discursively (Bazerman 1999).

To explore some of the ways in which AI systems are scaffolded discursively, I analyse a set of research articles about a conversational software agent system called REA. The designers describe REA in their research articles as human, or very nearly so. The system is presumed repeatedly to be “represented as a human” and “as a person” (Cassell 2001), while interactions with users are repeatedly described as “natural.” Other rhetorical strategies reinforce this claim to REA’s humanness. The designers’ texts, for example, always use gendered pronouns as substitutes for REA, deflect attention away from REA’s artificiality and call attention to the system’s

¹ This metaphor does not guarantee that all computer systems will be interpreted as humanlike. Rather, the point here is simply that cyborg discourse is grounded on the potential identity of humans and machines. While it might be more precise, later in this paper, to rewrite this metaphor as A Software Agent is A Human, it is in fact my contention that cyborg discourse contributes to the blurring of the boundary between humans and machines, and thus opens up an interpretive space in which it makes sense (or at least it does not seem so bizarre) to speak of humanlike computers.

“embodiment”, position REA as a discursive subject, obscure designers from the text (Hodge and Kress 1993), invoke the trope of the scripted drama when representing interactions between REA and users, project an entity structure and boundaries on the system (Lakoff and Johnson 1980), distinguish between the humanlike figure on the screen and other system components which are discursively constructed as “her” possessions, and visually construct REA as human by depicting interactions with users as unmediated by electronic equipment (such as a headset microphone). These rhetorical strategies systematically downplay REA’s artificiality by positioning the system linguistically as an autonomous, gendered agent. In the final section of this paper, I argue that, within the context of the dream of AI to produce humanlike machines (Baggi 2000; Negrotti 2000), and given our strong bias for human-human interaction (Picard 1997), the presumption of REA’s humanness in the designers’ research articles needs little justification.

The agent paradigm in AI

The AI marketplace of the early 1980s, according to Janet Vaux (2001), was informed by a “cyborg discourse” in which inscrutably smart “expert systems” held sway over the dissemination and production of knowledge. This discourse was grounded on a perceived identity between humans and machines such that human intelligence could be defined in terms of (or reduced to) the ability to manipulate symbols. Computers “are capable of thought” because, according to the “strong symbol system hypothesis” (Copeland 1993) that informs this view (also called symbolic AI), “at bottom all human thinking consists of symbol manipulation.” While Vaux (2001) identifies a shift in the mid- to late-1980s away from cyborg discourse within the marketplace of AI vendors, the academic arm of AI continues to shape and be shaped by the cyborg dream to reproduce human nature in a machine. According to Eugene Charniak and Drew McDermott, in their *Introduction to Artificial Intelligence* (1985, quoted in Bringsjord et al. 2003), “The ultimate goal of AI (which we are very far from achieving) is to build a person, or, more humbly, an animal.” More recently, Denis Baggi (2000) has echoed a similar sentiment: AI “represents the realisation of the ancestral dream, that of building the *homunculus* or thinking machine.” Thus, AI might be considered a master discipline, “the experimental bench for exploring all hypotheses,” to which all other disciplines (e.g., philosophy, psychology, linguistics, anthropology, biology, neurophysiology, etc.) are subsumed in the pursuit of the “goal of reproducing something which is, properly or not, referred to as *natural*.”

In the field of AI, this goal has been reflected most clearly, especially since the early 1990s, in the coalescing figure of the *intelligent agent*. According to Bringsjord et al. (2003), “all of AI has been to an astonishing degree unified around . . . the conception of an intelligent agent.” The unifying impact of the figure of the *agent* on the field of AI can not be overstated, although it “has in large part come courtesy of a comprehensive textbook intended to cover literally *all* of AI: Russell and Norvig’s (1994) *Artificial Intelligence: A Modern Approach (AIMA)*, the cover of which also displays the phrase ‘The Intelligent Agent Book’” (Bringsjord et al. 2003). According to one of its authors (Russell), *AIMA* is used in over 600 schools for 738 courses, including “*all* the schools in the top 30 that have an AI course” (Russell 2002). The figure of the *agent* serves as a way of directing AI researchers’ attention by framing problems and solutions in terms of autonomous, social entities/agents (as opposed to objects or modules as in the paradigm of object-oriented programming) that possess mental states comprised of beliefs, goals, knowledge, obligations and desires. Agents are represented as social, cooperating entities that must coordinate their actions with other agents in order to accomplish specific goals in an environment that is “physical and partially unpredictable” (Shoham 1993). In other words, it is usually assumed, within the agent paradigm, that no single agent possesses a thorough understanding of its environment or the means to achieve its goals on its own.

Yoav Shoham (1993) distinguishes between object-oriented programming (OOP) and agent-oriented programming (AOP):

Intuitively, whereas OOP proposes viewing a computational system as made up of modules that are able to communicate with one another and that have individual ways of handling incoming messages, AOP specialises the framework by fixing the state (now called the *mental state*) of the modules (now called *agents*) to consist of components such as beliefs (including beliefs about the world, about themselves and about one another), capabilities, and decisions, each of which enjoys a precisely defined syntax. Various constraints are placed on the mental state of an agent, which roughly correspond to constraints on their common sense counterparts. A computation consists of these agents informing, requesting, offering, accepting, rejecting, competing, and assisting one another.

Shoham (1993) establishes a parity between human agents and software agents by suggesting that, despite the differences between humans and machines, the actions of intelligent agents might be described in terms of human actions and behaviour, what Shoham calls “mental terminology.” The mental states of humans and software agents are presumably equivalent—or can at least be

treated as equivalent within the agent paradigm—given that the “constraints” placed on the former “roughly correspond” to the constraints placed on the latter. In one of the more radical formulations of this view, Marvin Minsky (1988), co-founder of the AI Lab at MIT, has argued that the human mind itself is comprised of a large number of interconnected parts (agents) that form societies or agencies to accomplish highly specialised tasks.

Steven Johnson (1997) draws a basic distinction between personified (visible) and invisible agents. An example of a personified agent is a “digital valet” who manages your affairs for you. “In 1989, Apple released a celebrated video entitled *The Knowledge Navigator*, which deposited a genteel, tuxedoed actor in the upper-right-hand corner of a PowerBook.” The video, set in the year 2010, features a human professor of earth sciences (or some related field) who draws upon the affordances of a virtual butler to assist him with his academic research, manage his appointments, contact his colleagues, and so on. Not all agents are personified, of course. “Some of the most promising agents toil almost entirely behind the scenes” (Johnson 1997). An invisible agent, for example, may do your bidding by roaming the Internet for specific information: stock quotes, news articles, the best price on airfare, etc. The user may interface with this agent via a Web browser or other program that provides a set of controls for conveying the user’s preferences to the agent, but the agent itself functions independently of direct manipulation as it “travels” the network. Johnson (1997) distinguishes “travelling” agents from “personal” agents that “settle into your computer’s hard drive and stay there for good, watching your behaviour and helping out when they get a chance.” The third class of agent-driven interface, according to Johnson, is comprised of “social” agents that “compile relevant data for you by chatting with other agents, swapping stories and recommendations.”

Software agents may turn out to have “profound effects” precisely because they are the “most independent—the most autonomous—tools in the interface repertoire” (Johnson 1997). J. Macgregor Wise (1998) worries, for example, that “as agents become more efficient and fade into the woodwork, we will lose the opportunity to interrogate some of the social actors making the greatest impact on our lives.” According to Wise (1998), when tasks get delegated to technologies, humans have a tendency to forget about them:

We *delegate* tasks to technologies (traffic lights take the place of police officers standing at intersections; automatic springs on doors close them behind us so that we don’t have to, and so on). But as technologies take tasks off our hands, they also take them off our minds. Technologies perform not only so that we don’t

have to, but also so that we don't have to think about them (i.e., closing doors) any more.

When we delegate tasks to autonomous agents, we threaten to obscure the values and perspectives that inform their design and use. Technologies are not neutral, according to this view, but are inscribed with a set of assumptions that can “configure the user” in ways that can easily slip under our radar (see Woolgar 1991). For example, a seemingly simple and straightforward technology such as a hydraulic door-closer (Latour 1988) can “prescribe” a set of behaviours back onto humans and thus “discriminate” (see Winner 1985) against those potential users who do not meet the basic criteria for use (i.e., that the user should be ambulatory, of a certain height and strength, etc.). For Latour (1988), then, the machine is a text that is “inscribed” with assumptions about who should use it and how it should be used. Likewise, “the text itself is a machine” that, to borrow now from N. Katherine Hayles (1999), can reflexively produce the very objects (e.g., AI systems) whose existence it presupposes.² To view the text as a machine is to assume, in the case of AI, that research articles do not merely reflect AI systems that presumably exist outside the reach of discourse, but rather provide the necessary means of articulating and bringing such systems into being. As a framework for studying the social effects of AI, Latour's (1988) vocabulary of *inscription* (script, description, inscription, prescription, etc.) thus opens up a space for exploring textual practices in AI and their effects on our understanding of how software agents, whether personified or invisible, are “inscribed” in texts. In the following section, I describe discourse analysis as a methodology for studying the inscription of AI systems in research articles.

In this section, I have described the *intelligent agent* as a coalescing, unifying figure in AI since at least the early 1990s. By establishing a parity between human agents and software agents, the agent paradigm reflects the master dream of AI to recreate (human) nature “through alternate construction strategies as compared to the natural ones” (Negrotti 2000). Shoham's (1993) “mental terminology,” for example, establishes a rough correspondence between humans and machines

² I have appropriated Hayles' (1999) language here with the exception of one important substitution. Hayles (1999) suggests, as a way to explain the concept of reflexivity, that the US Constitution “is shown to produce the very people whose existence it presupposes.” The text of the Constitution does not so much describe the “people” as bring them into being in the very act of labelling/describing them. “Reflexivity is the movement whereby that which has been used to generate a system is made, through a changed perspective, to become part of the system it generates”.

by describing the latter in terms of the former. Apple's *The Knowledge Navigator* video is an example (albeit a more popular one) of how cyborg discourse informs our understanding of AI, insofar as it works out the explicit assumption in cyborg discourse that humans and machines are potentially equal.

Discourse and technology

Discourse analysis (DA) is a methodology that, when applied to the study of technological systems, directs our attention to the myriad ways in which technologies are “scaffolded” by discursive practices (see Gee 1999). According to this view, technology does not refer unproblematically to a set of artefacts whose value and meaning are plainly obvious to anyone. Instead, what counts as “technology” in the first place, not to mention how technologies come to be represented as (humanlike) entities (see Lakoff and Johnson 1980), are matters to be worked out as part of an ongoing social practice (see Blommaert and Bulcaen 2000; Fairclough and Wodak 1997). This view does not deny the materiality of technology or imply that technology can be reduced to language or mere rhetoric. But it does suggest that technology and discourse, in keeping with a materialist-constructionist perspective (O'Brien 1999; Wakeford 1999), are mutually constitutive. A discourse analytic perspective thus refuses to make hard and fast distinctions between technology and discourse, but assumes, in the case of personified agent systems, that technological artefacts are mediated discursively and aims to delineate the discursive dimensions through which they come to be apprehensible as humanlike entities. To study AI from a discourse analytic perspective is to attend to the relationships among discursive practices and materials and meanings that facilitate the doing of technology (Wakeford 1999).

Discourse analysis as a methodology for studying technology foregrounds the *materiality* of language to scaffold artefacts such as AI systems, and thus problematises the “conduit” metaphor in which language is assumed to provide a clear, straightforward, transparent means of transferring information from one mind to another (Lakoff and Johnson 1980). Some problems with the conduit model are that it assumes that language is immaterial and purely descriptive, that language exists independently of speakers and hearers and that language merely reflects objects in the world without playing an important role in the production and maintenance of that world and the objects within it. The DA view assumes, in contrast, that language gives meaning, value and form to AI systems. Language and texts are the objects of interest in DA and not merely resources for explaining or describing objects and ideas that presumably exist outside the reach of language and

discursive interaction (see Wood and Kroger 2000). Even seemingly referential or neutral texts, such as traditional scientific reports and conference papers, do not exhibit “the absence of a position,” according to John Flowerdew (1999), “but one position among others.” By assuming that the “scientific report . . . serves to mask the unsystematic and human aspects of the research process as it actually is”, the discourse analyst recasts objectivity as a rhetorical achievement.

To focus on texts and discourse rather than authors and designers is to imply that the type of analysis of written texts I have been describing is concerned with visible inscriptions (marks on the page) and not with what a writer might have been thinking or doing while writing. The question of whether a writer/designer actually *intended*, consciously and wilfully, to express herself in a particular way in a text is simply not the focus of this analysis. In fact, the view espoused here is that writers are never in complete control of the meaning of their texts anyway, since those texts will undoubtedly be consumed and interpreted by readers with differing motivations and experiences.

Our understanding of AI is channelled through particular configurations of discourse. One such configuration, I have suggested, is “cyborg discourse,” which is informed by the belief that humans and machines are potentially equal. The driving metaphor in cyborg discourse (and in the agent paradigm, I would argue) is: A Computer Is A Human. This metaphor provides a starting place for exploring some of the ways in which texts construct intelligent agents as human. It is closely related to another metaphor, An Instrument Is A Companion (Lakoff and Johnson 1980), because both metaphors call attention to our tendency “to treat certain significant instruments like cars and guns as companions, giving them names, talking to them, etc.” As I have argued in this section, texts do not merely reflect or describe a set of pre-discursive facts about a particular software agent. Rather, the development of a technology should be understood within a discursive system that gives it “presence, meaning, and value” (Bazerman 1999).

Disembodied conversational agents

The Real Estate Agent (REA) is a software agent currently under development at the Gesture and Narrative Language (GNL) research group, which is part of MIT’s Media Lab. The GNL group is under the direction of associate professor Justine Cassell, and currently consists of eight graduate students, eight undergraduate students, and three non-faculty researchers (GNL Website 2002).

The REA agent is the main focus of at least fifteen of the group's published articles between 1999 to 2002, and it is this set of texts that I analyse in this paper.³

According to Cassell et al.,⁴ REA is designed to “enable and enhance natural forms of communication and linguistic expression” (MIT Media Lab Website 2002) by capitalising on the ways in which humans communicate verbally and non-verbally with other humans in face-to-face settings. The system is designed to “make interactions almost as natural as face-to-face conversation with another person. In the current task domain, REA acts as a real estate salesperson, answering user questions about properties in her database and showing users around the virtual houses” (Cassell et al. 2000b). Cassell et al. (2000a) explain that they “chose the domain of real estate because of the amount of talk with a social function that it provokes... This domain, then, allowed us to work toward an agent capable of dealing with multiple communicative goals: both interactional and propositional contributions to the discourse, made by way of conversational behaviours carried out in several modalities” (Cassell et al. 2000a). Several computers are used to provide the agent with multi-modal, humanlike capabilities. A user interacts with the REA system by standing in front of a large projection screen, and communicating (speaking, gazing, gesturing, moving) “naturally.” REA is figured as a young white woman, in professional attire, with short (or pulled-back) hair topped with sunglasses. Approximately three-fourths of the figure's “body” (head to hip) is visible on the projection screen. Behind the figure is displayed a furnished room or an entire building from the system's database of real estate properties in the Boston area.

³ While the GNL group has written at least seventeen papers about REA to date, two of these papers were not counted in my sample because they are basically identical to two other papers in my sample. All of GNL's most recent publications and conference papers are available online. Page references in the text refer to page numbers in the electronic copies of GNL articles, not to page numbers in the journals and proceedings where some of the articles have been published. Download articles from <http://web.media.mit.edu/~justine/publications.html> and <http://gn.www.media.mit.edu/groups/gn/publications.html>.

⁴ I often use “Cassell et al.” as shorthand for all of the members of the “Gesture and Narrative Language” group, even when a specific article I am citing has not been first-authored by Cassell. I do this because I am assuming that Cassell is the driving force of the group—GNL is *her* group after all—and she thus plays a critical role in all the group's publications, even in rare cases where a GNL publication is not first-authored or single-authored by Cassell.

Interactions are almost always represented visually in GNL texts as some variation of Fig. 1. The artificiality of the interaction in this example is highlighted by the prominence of 1) the user's headset microphone and 2) the projection screen that separates user from system and reveals the limits of the digital medium. But this is not the only way in which human-computer interactions are visually constructed in GNL texts. For example, Fig. 2 downplays the system's artificiality by obscuring how interactions with the system are always mediated by technology (i.e., a projection screen, a headset microphone, several computers, etc.). In Fig. 2, the user is depicted as more fully immersed in the system: the headset microphone and screen border are gone. The result is the visual construction of a locus of interaction that seems more natural, immediate, and "transparent" (see Bolter and Grusin 1996). In short, the interaction is mediated in Fig. 2 by a visual rhetoric that obscures its very artificiality.



Fig. 1. A user stands in front of the REA interface while wearing a headset microphone; the edge of the projection screen is visible at top right. This is the standard way that GNL texts depict interactions visually (Source: Cassell et al. 2000a). Used with permission of the MIT Press



Fig. 2. A user stands in front of the REA interface. Headset microphone and screen border are not visible. The result is an interaction that appears to be unmediated (Source: Cassell 2001, p.74). Copyright© 2001, American Association for Artificial Intelligence

What is remarkable about the REA system is that it reveals the extent to which members of the AI community are seriously pursuing the dream of AI to recreate human nature by artificial means (see Baggi 2000). REA is the product of such a commitment to build a reliable cadre of quasi-humans. But REA's humanness can also be interpreted as a rhetorical achievement. As an example of what Cassell and others (see Cassell and Bickmore 2002) are calling embodied conversational agents (ECAs)—i.e., software agents with bodies and verbal and nonverbal conversational abilities—REA does not have a physical body, not even a robotic one. The system's embodiment is entirely virtual, two-dimensional and digitally confined to the boundaries of the projection screen. Nevertheless, REA's virtual body is described in ways that produce the system *as human*. For example, a number of GNL articles appeal to the affordances of the “human body” in the design of humanlike software systems (Cassell et al. 2000b), equate REA's verbal and nonverbal behaviour with “natural, full-bodied (in some sense) conversation” (Cassell and Bickmore 2002), and make the claim that REA is “represented as a human” and “as a person” (Cassell 2001). The GNL Web page about REA, moreover, boldly describes the system as “human in form”:

We are developing autonomous agents that are capable of having a real-time face-to-face conversation with a human. These agents are human in form and communicate using both verbal and non-verbal modalities. We believe that such agents provide a new form of human-computer interface which users can interact with naturally, without training, since they already know how to engage in face-to-face conversation with other people.

In fact, the GNL group seems to demand of us that we evaluate REA according to whether the system acts, talks and looks human. We are encouraged, throughout the GNL texts, to equate REA with a human woman, and thus the strength of the group's argument (and hence the usefulness and viability of its REA system) seems to depend upon whether REA meets our expectations as a near-perfect reproduction of nature, of embodied humanness.

To insist on REA's presumed physicality in the GNL texts is to deflect attention from the artificiality of REA's (virtual) embodiment. The appeal to REA's humanlike body supports GNL's argument that humanlike agents will promote “natural” interaction between computers and humans. An astute observer might qualify this appeal by pointing to the differences between human (physical)

embodiment and REA's virtual embodiment. But this sort of clarification never makes its way into the GNL texts because the practicality of GNL's research agenda depends upon a discursive strategy in which REA's virtual, animated body is presented as identical to the user's human body. If REA were to be described in the GNL texts as machine-like, as a cartoon figure, or as severely limited by the constraints of the system's digital medium/screen, then the designers' overall argument in support of ECAs would suffer. But given the strength with which Cassell et al. discursively appeal to the system's human form, the group's texts never raise the question of whether the system's embodiment might in fact be dis-embodied, virtual, or un-humanlike. Instead, the texts function rhetorically to bolster GNL's main justification (i.e., in terms of "natural," untrained interaction) for the design of ECAs.

This appeal to REA's human body does not need to be supported, let alone questioned as problematic, because it makes sense to us within the context of our human tendency to interact with nonhuman objects (such as computers) in humanlike ways (Nass et al. 1994; Picard 1997). Provided certain constraints are met, the presumption of the other's humanness is offered readily by humans who chat with conversational machines wittingly or unwittingly. According to Internet lore, for example, many humans in text-based chat environments (e.g., MUDs, chat rooms and IRC channels) have mistaken software programs called "bots" (and sometimes called software agents) for human because they have assumed, despite being unable to see faces and bodies in text-based interactions, that all participants in the chat room are human. Perhaps the most well known bot, after ELIZA (see below), is JULIA (Mauldin 1994), which was designed in 1990 to log on as a regular player to a type of multi-user, text-based environment called TINYMUD. The JULIA program chats with other (human) players and provides MUD services such as messaging and help (see Zdenek 1999). Michael Mauldin (1994), the program's designer, tells a story (quoted in Foner 1993) of how one unsuspecting human called Barry "spent 13 days trying to get to first base with Julia, and it's not clear he ever figured out he was trying to pick up a robot (although he clearly suspects it at two points)." In the following excerpt, Barry (who has been given the nicknames of "cujmeister" and "cujster" by the program) questions JULIA's nonhuman behaviour, but not in any sustained, serious manner:⁵

barry says, "hello, I am barry. How are you today"

Julia says, "Hello, cujmeister."

⁵ Excerpts are shown here as they originally appeared in the sources: no changes to spelling, punctuation, or grammar have been made.

barry says, "The Jularama, Ha ya doin? What state are you in"

Julia says, "I never really thought about it, barry."

barry says, "What do you look like"

Julia says, "I'm 5'9" tall, weigh 149 lbs, with medium-length platinum blond hair and green eyes."

barry says, "Howdid you answer so fast"

Julia says, "So?"

barry says, "Oh, Iam looking for someone to share secrets with"

barry says, "hello...hello...are you with me"

Julia says, "Hello, kujster."

barry says, "hey woman how can I eat at the café"

Julia says, "I'm sorry I can't help you, my map doesn't seem to work, barry."

Barry's presumption of JULIA's humanness greatly outweighs any suspicion he might harbour about the program's nonhuman status (i.e., it types so fast that it must be a machine). In fact, Barry does not appear to be troubled by other indications of the program's machine-like behaviour in this excerpt, such as when the program fails to respond ("hello...hello...are you with me") or outputs an irrelevant or "last resort" reply ("my map doesn't seem to work," "I never really thought about it").

Leonard Foner (1993) associates Barry with one of those "unlucky and clueless players [who] can be led seriously astray." But I would suggest that Barry is only doing what comes naturally to humans, even to humans who admit to knowing the real identity of their (nonhuman) interlocutors. Consider, for example, how some users have reacted to ELIZA, a well-known chat program designed by Joseph Weizenbaum (1966) in the mid-1960s. The program relies on very simple but powerful techniques for simulating humanness, namely pattern matching and input sentence transformation (see Copeland 1993). Weizenbaum also "loaded" the context of interaction in such a way that users are encouraged to attribute intelligence and humanness to what is only a very simple program. In the case of ELIZA, the program's simulation of a therapist in the Rogerian, non-directive (know-nothing) style promotes users' attributions of intelligence to the machine, because ELIZA, which has no knowledge base or reasoning ability, is "free to assume the pose of knowing almost nothing of the real world" (Weizenbaum quoted in Copeland 1993). (Text-based online interactions with bots are also *context-loaded* because users cannot see the bodies and faces of other participants; the same programs would not pass for human in a more demanding, face-to-face context.) Despite ELIZA's simplicity, however, many users have "form[ed] a strong

emotional bond with ELIZA—a bond that Weizenbaum found obscenely misplaced” (Copeland 1993). In one case, Weizenbaum’s secretary—“who on one level knew very well that Weizenbaum’s creation was just a machine—took to insisting that other staff leave the room so she and ELIZA could talk in private” (Copeland 1993). In another case, some psychiatrists were interested in using the program to solve a shortage in human therapists: “Several hundred patients an hour could be handled by a computer system designed for this purpose” (Colby et al. 1966, quoted in Copeland 1993).

I share Weizenbaum’s sense of horror at the thought of human therapists being replaced by a software program that, according to Jack Copeland (1993), “knows nothing about her environment, has no reasoning ability, cannot plan out actions, cannot learn anything, and possesses no understanding of her own motives.” That a simple software program can evoke feelings of intimacy and trust from users is a testament to the social aspects of human-machine interaction. (Interestingly, while Copeland says that the ELIZA program is not a thinking, intelligent entity, his use of gendered pronouns and reliance on a vocabulary of human actions to describe the program imply, paradoxically, that ELIZA is like him—intelligent, human, and alive.) The sociality of human-machine interaction has been well-documented by Clifford Nass and his colleagues at Stanford (e.g., Nass et al. 1994; Reeves and Nass 1996), who have shown in numerous studies how “users can be induced to elicit a wide range of social behaviours, even though users know that the machines *do not* actually possess feelings, ‘selves,’ genders, or human motivations” (Nass et al. 1994). Upon reviewing the findings of Nass et al., Rosalind Picard (1997) concludes that:

It is not unusual for intelligent people to attribute emotion to things that clearly do not have emotion. For example, someone might wind up a toy dog to make it wag its tail, and say, “How cute—it likes us.” Although people *know* that wind-up toys and computers do not have emotions, their default model for relating to others apparently assumes them, most likely because humans are strongly biased for human-human interaction.

This strong bias for human-human interaction, even when one party knows the other to be a machine, provides an answer to the question of why people like Barry hold on to their belief in JULIA’s humanness despite evidence to the contrary, or why some users of ELIZA insist on treating the program as human despite knowing that it is only a simple program.

Where does this bias come from? I would suggest, drawing on George Lakoff and Mark Johnson (1980), that “our conceptual system is grounded in our experiences in the world”:

We are physical beings, bounded and set off from the rest of the world by the surface of our skins, and we experience the rest of the world as outside us. Each of us is a container, with a bounding surface and an in-out orientation. We project our own in-out orientation onto other physical objects that are bounded by surfaces. Thus we also view them as containers with an inside and an outside...But even where there is no natural physical boundary that can be viewed as defining a container, we impose boundaries—marking off territory so that it has an inside and a bounding surface.

We draw on our experiences as social beings to make sense of concepts that “are not clearly enough delineated in their own terms to satisfy the purposes of our day-to-day functioning” (Lakoff and Johnson 1980). Concepts that are “less concrete or less clearly delineated”—e.g. “Love, Time, Ideas, Understanding, Arguments, Labour, Happiness, Health, Control, Status, Morality, etc.”—are metaphorically defined in terms of concepts that are familiar to us, concepts that are grounded in our experience. Given our propensity 1) to “project a front-back orientation in context onto objects that have no intrinsic fronts or backs” ; 2) to treat objects as “entities” by “often projecting boundaries and surfaces on them where no clear-cut boundaries or surfaces exist naturally” ; 3) to view instruments as companions; and 4) to presume, according to an anthropocentric view (see Kiriazis and Slobodchikoff 1997), that any entity that appears to possess even the most basic capacity for language must be human, is it no wonder that software systems, especially systems designed and marketed as “intelligent agents” (Wise 1998), are often metaphorically compared to human beings?

Even simple software programs like ELIZA are not clearly delineated, since the typical user can not readily see where the program’s boundaries are or how complex it is “under the surface.” In the case of more complex systems like REA, I would suggest that it is impossible for any single person to conceive the totality of the system without imposing some kind of order on it, containing it with boundaries, projecting an entity structure on it, and comparing it metaphorically to what is familiar to us as human beings. The REA system, for example, consists of a highly complex network of software systems, computers, and other hardware devices such as cameras and a microphone:

The system currently consists of a large back projection screen on which REA is displayed and in front of which the user stands. Two cameras mounted on top of the projection screen track the user’s head and hand positions in space. Users wear a microphone for capturing speech input. A single SGI Octane computer runs the

conversation engine (originally written in C++ and CLIPS, currently moving to Java), while several other computers manage the speech recognition (until recently IBM Via Voice; currently moving to SUMMIT) and generation (previously Microsoft Whisper; currently BT Festival), image processing (STIVE) and VGUS and graphics (written in OpenInventor). (Cassell 2001)

The system's overwhelming complexity is greatly reduced, however, when the system is inscribed rhetorically as a humanlike subject. As a bounded, apprehensible entity, REA thus becomes identified with the lifelike figure on the projection screen, but not with the user's microphone or the database of real estate properties:

REA's domain of expertise is real estate; she has access to a database of available condominiums and houses for sale in Boston. She can display pictures of these properties and their various rooms and point out and discuss their salient features. (Cassell 2000)

When REA is figured as a humanlike entity, the database of homes can be said to belong to "her," rather than comprising part of the overall system. REA is thus defined as an autonomous entity "who" possesses/uses other system components (a domain of expertise, a database, pictures, salient features). These other components, however, might not appear to be so easily separable from "her" if they were to be described at what Negrotti (2000) calls a different "observation level." A different observation level, for example, might call attention to how the system's various components are managed as a function of the billions of instructions processed by the system's several computers in one second. While Cassell et al. do in fact offer descriptions of the system that call attention to its various components (as in the first example above), they more often describe the system as an autonomous, bounded entity. Their texts regularly impose order on REA, define it in terms of what is familiar to us, and thus inscribe it with an apprehensible identity. To do otherwise would be to threaten GNL's research agenda by deconstructing or dispersing the lifelike entity ("she") who presumably manages the system. Whereas a description of the system at the level of lifelike figure on the projection screen named REA divides the system up according to "her" possessions, an observation level that describes REA as a complex network of computers, other devices such as projection screen, cameras and microphone, as well as numerous software systems, would structure our understanding of the REA system very differently. In short, Cassell et al. make their case by downplaying descriptions of REA as a distributed system in favour of descriptions of REA as a human.

To enforce the argument that REA is humanlike, Cassell et al. always gender REA in the texts and most always represent it discursively as an autonomous entity whose relationship to its designers is tenuous at best. Often, the designers are absent altogether from descriptions of the system, “deleted” from the text and replaced by REA, “the nominalised action” (Hodge and Kress 1993):

REA simultaneously processes the organisation of conversation and its content.

When the user makes cues typically associated with turn taking behaviour such as gesturing, REA allows herself to be interrupted, and then takes the turn again when she is able. (Bickmore and Cassell 2001)

That REA “allows herself” to do anything at all implies that the system possesses an internal mind, that at some level the system is uncontrollable, out of human reach, that it is not simply an inert object (or very large set of instructions) to be manipulated directly (see Zdenek 2002). While I do not doubt that the REA system is lifelike and complex, I am interested in the rhetorical functions of descriptions like this one to make the designers’ case for them. In other words, I would suggest that by gendering REA by default, positioning the system as a discursive subject “who” performs actions independently, and deleting themselves from the text, Cassell et al. produce the system *as human* rather than simply describe it as such. To demonstrate how this might be so, we would simply need to imagine other ways of describing the system that are currently subordinate to GNL’s dominant strategy of presenting the system as an autonomous, humanlike entity. For example, one alternative would foreground the designers in the text and thus disperse the system’s autonomy over many components; another way would describe REA’s behaviour not as the result of “her” internal mind per se but as the result of the coordination of a complex network of systems and instructions.

REA is also produced as a human agent via the trope of the scripted drama. GNL texts always represent extended interactions between users and REA in terms of the conventions of a dramatic dialogue, as in the following exchange between REA and a human user named Mike (Cassell 2000):

Mike approaches the projection screen while Rea is gazing about idly. As Mike moves within range of the two cameras mounted above the screen, Rea turns to face him and says:

Hello. How can I help you?

Mike. I’m looking to buy a place near MIT.

Rea nods, indicating that she is following.

Rea. I have a house to show you.

A picture of a house appears on the screen behind Rea, who blinks and looks at the house and then at Mike.

Rea. It's in Somerville.

Mike. Tell me about it.

Rea looks up and away while she plans what to say.

Rea makes an expansive gesture with her hands.

Rea. It's big.

Mike brings his hands up as if he is about to speak, so Rea does not continue, waiting for him to speak.

Mike. Tell me more about it

Rea. Sure thing. It has a nice garden.

Rea sketches a curved gesture with her hands, indicating that the garden extends along two sides of the house.

By representing interactions in terms of a scripted drama, Cassell et al. are not simply presenting an unbiased or neutral report of what has transpired between a user and a machine, but constructing, in writing, how machines like REA are human. In this example, Cassell's description of what REA is doing is informed by the larger purpose of the GNL texts, which is to show how GNL is designing software agents that communicate "naturally." REA is described in terms of human action (gazing, blinking, planning, waiting, etc.) because the main goal of the article is to show how REA is humanlike. Note that this does not necessarily imply that the system is inherently humanlike. Rather, my point is that the descriptions themselves help constitute the meaning of technologies like REA. Technologies do not begin and end where their programming code and hardware begin and end, but are inscribed in texts just as surely as they are symbolically inscribed in silicon. By simulating the conventions of a drama, designers like Cassell et al. benefit from the assumption that speaking parts in a drama are always played by humans: REA is humanlike not simply because the program is inherently humanlike, but because it is described within a (dramatic) context in which it makes sense to think of it as such.

Defending nature

Given our human tendency to inscribe systems such as REA with agency and autonomy—i.e., to treat instruments as companions—much of the case for the value and feasibility of ECAs has already been made, I would argue, *prior* to the reader's engagement with a GNL text. The argument

in support of software systems that users can interact with “naturally” is so intuitively strong that little defence is usually required or offered. Why might this be? Because the GNL texts are informed by the assumption—described above as key to understanding “cyborg discourse”—that humans and machines are potentially equal.

The basic GNL argument that theories of human-human interaction might provide the basis for an understanding of human-computer interaction is never rigorously defended in the GNL texts. The group’s repeated appeal to the identity of human-machine and human-human communication is treated as self-evident and obvious. In fact, only two justifications are regularly offered in support of the argument that human-machine interaction ought to be modelled on human-human interaction. Accordingly, 1) people treat computers as social actors, and 2) people find humanlike (i.e., animated, embodied, conversational) computer systems to be more engaging, helpful, satisfying, etc. For support, Cassell et al. repeatedly appeal to a) research in the “Computers are Social Actors” (CASA) paradigm, and to b) other studies by AI researchers on the effects of personification on users of animated interfaces.

As for the first appeal, Cassell et al. invoke the work of Nass and his colleagues (see above). Nass, Steuer and Tauber (1994) have developed a five-step approach to testing whether CASA is true:

1. Pick a social science finding (theory and method) which concerns behaviour or attitude toward humans. The studies presented here draw from social psychology and sociology.
2. Change “human” to “computer” in the statement of the theory.
3. Replace one or more humans with computers in the method of the study.
4. Provide the computer with characteristics associated with humans: (a) language output; (b) responses based on multiple prior inputs; (c) the filling of roles traditionally filled by humans; and (d) the production of human-sounding voices.
5. Determine if the social rule still applies.

Given that Cassell et al. are interested in the potential identity of human-human communication and human-computer interaction, CASA provides them with a straight-forward justificatory framework. Among other things, the CASA paradigm (Reeves and Nass 1996) reveals how users may rely on gender and racial stereotypes to make sense of computers that are designed to act, talk, or look in explicitly gendered ways. CASA also shows how “users react differentially to social agents based on their own personality” (Bickmore and Cassell 2001).

When citing CASA literature, Cassell et al. often rely on a simplified version of CASA which serves to make it sympathetic to their research agenda. Twelve of fifteen articles in my sample cite one or more studies in the CASA literature. The following example is an exception, I would suggest, because it goes beyond a basic, one-sentence summary of CASA:

In a series of studies in the “Computers As Social Actors” paradigm, researchers have demonstrated the possibility of manipulating the user’s relationship with a computer using a wide range of behaviours. Reeves and Nass demonstrated that users like computers more when the computer flatters them. Morkes, Kernal and Nass demonstrated that computer agents which use humour are rated as more likable, competent and cooperative than those that do not. Moon demonstrated that a computer which uses a strategy of reciprocal, deepening self-disclosure in its (text-based) conversation with the user will cause the user to rate it as more attractive, divulge more intimate information and become more likely to buy a product from the computer. (Bickmore & Cassell 2001)

This example is atypical. Cassell et al. more often appeal to a simplified version of CASA. This version is more sympathetic to their research agenda because it neatly sidesteps any discussion of the technical, not to mention practical and social, complexities involved in designing humanlike machines. The following examples feature more typical appropriations of the CASA paradigm in the GNL texts:

One way for a software agent to overcome the credibility problem is to embody the agent in the interface, that is, provide it with a human appearance and lifelike behaviour. It has been shown that people tend to treat computers as social agents anyway (Reeves and Nass 1996), but providing an agent with an embodied presence provides a strong cue to users that the agent should be treated and interacted with as another person. (Bickmore and Cassell 1999)

Whether or not computers look human, people attribute to them human-like properties such as friendliness, or cooperativeness. An embodied conversational interface can take advantage of this and prompt the user to naturally engage the computer in human-like conversation. (Cassell et al. 1999)

The use of “anyway” in the first example suggests one way in which Cassell et al. invoke CASA to support their own research agenda. “Anyway” offers de facto justification for the design of embodied conversational software systems, but also smacks of a kind of anti-intellectualism, an eagerness to defer blindly to the masses who are *going to do it anyway, so why shouldn’t we help*

them do it more efficiently? The second example makes a similar argument because it implies that people will treat computers as human “whether or not computers look human.” Both examples suggest, paradoxically, that “natural” interaction does not come naturally, since it may need some encouragement or “prompting.”

In sum, the CASA paradigm provides Cassell et al. with a persuasive argument to support the design of software systems that exploit the so-called “natural” tendencies of humans to treat computers in social ways. When used in its most simple form (one or two sentences), CASA supports the contention that people everywhere regularly ascribe computers with social qualities. CASA thus justifies GNL’s attempt to build an “interactive system that allows humans to communicate *naturally*” (Cassell and Stone 1999), since people are going to interact with computers as social actors “anyway.” This connection is made explicit when Cassell and Stone (1999) appeal to “natural” communication in the context of Reeves and Nass’s (1996) argument that “humans will attribute social responses, behaviours, and internal states to computers.” Natural communication between humans and machines is thus explained in terms of the CASA phenomenon.

CASA can also be used, as the example above suggests, to support the second part of the justification for building agents that exploit “natural” communication, namely that people find animated agents more likeable, entertaining, educational, etc. To support this claim Cassell et al. often draw on a handful of studies by AI researchers on the effects of personification on users:

Prior research into embodied interface agents has found that users like them and find them engaging. In this paper, we argue that embodiment can serve an even stronger function if system designers use actual human conversational protocols.

(Cassell et al. 2000b)

This example comes from the first two sentences of the article’s abstract. The claim in sentence one is presented as a self-evident discovery (“Prior research . . . has found”), and provides a ground for warranting the claim made in sentence two. Cassell et al. thus imply that the personification question has already been answered. While they do in fact support this claim later in the same article by citing some of the research on personification, they are also very selective about what they cite, and typically only include research on personification that is sympathetic to their own view, including articles by Koda and Maes (1996), Takeuchi and Naito (1995), and Andre, Rist and Mueller (1998). Tomoko Koda and Pattie Maes (1996), for example, ask whether “adding faces to an interface” provides “better human-computer interactions.” They conducted an experiment in which subjects were asked to rank different interfaces (e.g., “no face, a female realistic face, a female caricature face, a dog’s caricature face, and a line-drawn smiley face”) on the basis of such

factors as likeability, intelligence, engagement and comfort level, and concluded that “[s]ubjects rated a realistic face as more intelligent, likable, and comfortable than a caricature face.” This finding supports what James Lester et al. (1997) call the “persona effect,” or the contention that “the very presence of an animated agent in an interactive learning environment—even one that is not expressive—can have a strong positive effect on student’s [sic] perception of their learning experience.” While Koda and Maes (1996) are not explicitly interested in the pedagogical implications of personification, their claim does implicitly support the persona effect. Based on their review of studies such Koda and Maes (1996), Cassell et al. imply that the persona effect is a given—the persona question has already been settled—that “the mere presence of a character wins us points” (Cassell et al. 2000b). Thus, they tend to emphasise those studies of personification that are sympathetic to their argument.

When Cassell et al. do cite research that is critical of the persona effect, they are careful to keep their own work out of the reach of the critique. For example, they cite a review by Doris Dehn and Susanne van Mulken (2000) in four articles in my sample that criticises Koda and Maes (1996) and questions the validity of the persona effect. (This same review, not surprisingly, is never cited by Cassell et al. together with proponents of the persona effect, such as Koda and Maes.) The Dehn and van Mulken review was accepted for publication in June 1999, published in January 2000 but read by one or more GNL group members prior to its publication.⁶

Based on a review of empirical research on animated interface agents, Dehn and van Mulken (2000) conclude that “[t]o date, the literature does not provide evidence for a so-called persona effect.” Cassell et al. acknowledge that, in light of Dehn and van Mulken’s (2000) review, the benefits of personified systems are “arguable,” but because the review locates the work of the GNL group as potentially outside the scope of the review, the review in fact serves not as a critique at all but as support for the GNL agenda:

⁶ In “Human Conversation as a System Framework,” Cassell et al. (2000a) cite Dehn and van Mulken in their references as “forthcoming.” It is difficult to say, however, just how many GNL articles in my sample had been published or presented prior to the availability among GNL group members of Dehn and van Mulken’s review. A conservative estimate would put the number in my sample at 12, since the remaining 3 of 15 articles were published in the fall of 1999, which is when GNL group members would have first read the Dehn and van Mulken review but not necessarily before they wrote the three 1999 articles in my sample.

To date, few researchers have empirically investigated embodied interfaces, and their results have been equivocal. As Shneiderman points out, there is ample historical evidence—in the form of a junk pile of abandoned anthropomorphic systems—against using anthropomorphised designs in interface design.

The recent evaluations of animated interface agents by Doris M. Dehn, a psychology professor at the University of Saarland in Germany, and Susanne van Mulken, a researcher at DFKI in Germany, concluded that the benefits of these systems are arguable in terms of user performance, engagement with the system and even attributions of intelligence. However, they also pointed out that virtually none of the evaluated systems managed to exploit the affordances, or special characteristics, of the virtual human bodies they inhabit. This design paradigm of embodying the interface “can only be expected to improve human-computer interaction if it shows some behaviour that is functional with regard to the system’s aim.” In other words, embodiment for the sake of pretty graphics alone is unlikely to work.

Note too that only recently have embodied conversational agents been implemented with anywhere near the range of conversational properties I’ve outlined here. For this reason, we are only beginning to be able to carry out rigorous evaluations of the benefits of conversational embodiment. In my research lab [...] (Cassell 2000)

This example is not a response to potential criticisms of animated software agents, but the construction of such criticism as outdated or irrelevant. In fact, within the context of the four GNL articles that cite it, the Dehn and van Mulken (2000) review should not be interpreted as a critique at all. Why? Because Dehn and van Mulken (2000) make a distinction between agents that exhibit verbal and non-verbal behaviour, locate Cassell and Stone (1999) as the “only study in which the animated agent showed non-verbal...behaviour”, and then praise research on non-verbal behaviour. The Dehn and van Mulken (2000) review thus serves, in the context of GNL’s appropriation, to distinguish between the earlier, failed attempts to build animated agents and the more successful work being undertaken in Cassell’s own “lab.” I would argue that the discussion of Dehn and van Mulken (2000) in the four GNL articles serves as a token gesture that satisfies the need, in academic writing, to respond to potential or actual criticisms of one’s research. The Shneiderman (1998) critique mentioned in the excerpt is more damning of animated interface research but also dispensed

with quickly and not a little unfairly by implying that it does not apply in the case of GNL's cutting-edge research.

In sum, Cassell et al. justify the design of software systems that elicit "natural" communication by invoking research on CASA and studies in AI on the persona effect. In the case of the CASA paradigm, Cassell et al. appeal to a simplified but powerful version that works to authorise their research agenda. If, according to GNL's argument, people tend to treat computers as social actors "anyway," then we might as well design computers that can exploit the implications of the CASA paradigm. In the case of the "persona effect," Cassell et al. frame the effect as a given, and not, for the most part, as an open question. This is probably because if they were to frame the "persona effect" as arguable, they would also call into question the foundation upon which their own research depends. While Cassell et al. cite research that questions the benefits of personified systems, their own research agenda, ironically, remains intact. What is "arguable" in their appropriation of Dehn and van Mulken (2000) is not the persona effect per se, but the persona effect in relation to all the previous research on animated interfaces that has preceded GNL's own. In citing Dehn and van Mulken (2000), Cassell et al. only give the appearance of responding to a criticism of their research.

This justificatory framework is not elaborate, as I have tried to show. A simple reference in an introduction to how users treat computers as social actors or to how users find animated agents likable and engaging is really all that is needed. The strongest justification, as I have suggested, is assumed and implied: no reasons are needed, within the context of the agent paradigm in AI, to justify a research agenda that is contributing to the realisation of the dream of AI to build the perfect humanoid, what Sengers (1998) calls the dream of the "robot friend." This appeal to the master narrative of AI—to the dream of all dreams—is often implicit in the GNL texts (first excerpt below) but may be explicit too (second excerpt below):

Our goal is embodied conversational agents that can engage humans in natural face-to-face conversation, including speech, nonverbal modalities such as gesture, gaze, intonation, body posture. (Cassell 2001)

One of the motivations for embodied conversational agents—as for dialogue systems before them—comes from increasing computational capacity in many objects and environments outside of the desktop computer—smart rooms and intelligent toys, in environments as diverse as a military battlefield or a children's museum—and for users as different from one another as we can imagine. *It is in part for this reason that we continue to pursue the dream of computers without*

keyboards that can accept natural untrained input. In situations such as these, we will need robustness in the face of noise, universality and intuitiveness, and a higher bandwidth than speech alone. We will need computers that untrained users can interact with naturally. And we believe that this naturalness of interaction can come from systems built on the basis of a strong model of human conversation.

(Cassell et al. 2000a, 60, first paragraph of conclusion, cited in full, italics mine)

The master narrative of AI (framed by GNL in terms of an assumed identity between human-human and human-machine communication) has immense rhetorical power, which is one of the reasons why it serves, in the articles that make up my sample, as a *frame* which can support a more or less technical discussion of embodied agent architectures or a theory of face-to-face interaction. For example, Cassell et al. often use introductions to refer to the presumed “naturalness” of software and hardware systems that exhibit humanlike properties. They also use conclusions (as in the examples above) to reiterate the argument that human-computer interaction can be as “natural” as human-human interaction. This appeal to “nature” serves to frame and cradle the “agentless” and “unnatural” descriptions of systems, components, agents, and interactions that often make up the architecture sections of these same articles. The nature frame also reinforces the presumption of REA’s humanness/naturalness that pervades these same texts as well.

Conclusions

In this paper, I have explored some of the ways in which AI is informed by the assumption in “cyborg discourse” that humans and machines are potentially identical. I have suggested that the agent paradigm can support a view of AI in which machines are described in terms of human action and “mental terminology” (Shoham 1993). This paradigm is scaffolded discursively, such that texts serve to rehearse and reinforce the assumption that machines can literally be human—or at least perceived as such when users are “prompted” by machines to interact “naturally” (see Cassell et al. 1999). According to this view, REA is not described as virtually human or as a mere digital representation of humanness so much as it is described as “human in form”. The GNL texts encourage readers to evaluate the system according to how well it talks, acts, and looks human. But to produce a machine that is humanlike, Cassell et al. must also construct REA discursively as human. The discursive production of REA as human is informed by a driving metaphor—A Computer Is A Human—that supports the underlying argument in these texts that machines and humans are potential the same. Within the context of the “ancestral dream” of AI to build a

“*homunculus* or thinking machine” (Baggi 2000), and given our strong bias for human-human interaction (Picard 1997), an approach to interface design that is presumed to be natural “anyway” (Bickmore and Cassell 1999)—i.e. presumed to be natural prior to the reader’s engagement with the text—does not necessarily need to be defended rigorously.

Of course, the texts analysed here tend to circulate among a very select group of readers, mainly among those colleagues or potential colleagues of Cassell et al. who regularly attend AI conferences (or aspire to do so) and keep up with academic journals in field such as AI, computer science and human-computer interaction. While I have argued that the identity of a computer system is always configured discursively, I have stopped short of suggesting that the identity of an agent system is not susceptible to being reconfigured by users and other “writers.” In other words, the identity constructed for REA by the GNL group, should the system become marketable in the near future, will necessarily have to compete with alternative interpretations (or ways of situating the system within specific contexts of use). Thus, while Cassell et al. evidently have invested significant discursive energy in producing REA as a humanlike woman, there is no guarantee that users will also appropriate their preferred reading should REA make it to the marketplace. To repeat, there is no way to guarantee that the marketplace of users will not subvert the key assumption in the agent paradigm that software agents are potentially human. At the same time, however, I would suggest, should REA move out of the lab and into the marketplace, that the system’s identity will continue to be a product of 1) our strong bias for human-human interaction, and 2) the weight of the discursive “load” (Latour 1991)—e.g., instruction manuals, training sessions, warning labels and other attempts to “*control* the relationship between reader and text [i.e., system]” (Woolgar 1991)—that will undoubtedly be placed on our interactions with the system by a team of designers that is heavily invested in the production of humanlike systems.

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